PYTHON FOR ACTUARIES · NOTEBOOK 06 · EXERCISE

Exploring Motor Insurance Claims

Hands-on Exercise

Explore `vehicle_fraud.csv` on your own with the tools from Notebooks 03, 04 and 05. Work through three phases:

- **Phase 1**: The Portfolio
- **Phase 2**: Claim Amounts
- **Phase 3**: Fraud

The hints name the tools; the code is up to you.

Several tasks repeat steps from the exercises in Notebooks 03 to 05 on purpose: here they come together in one exploration. Your code from there can serve as a starting point.

The Dataset: `vehicle_fraud.csv`

10,050 rows of synthetic motor insurance claims:

Column	Description
`claim_id`	Claim number
`policyholder_age`	Age in years
`occupation`	`Employed`, `Self-employed`, `Unemployed`, `Student`
`region`	Region `R01` to `R10`
`credit_score`	Credit score (300–850)
`customer_since_years`	Years as a customer
`vehicle_value`	Vehicle value
`vehicle_type`	`Compact`, `Mid-range`, `Luxury`, `SUV`, `Sports car`
`coverage_type`	`Liability`, `Partial coverage`, `Comprehensive`
`number_of_prior_claims`	Number of earlier claims
`claim_amount`	Claim amount
`police_report`	`1` if a police report was filed
`witnesses_present`	`1` if witnesses were present
`reporting_delay_days`	Days between accident and claim report
`fraud`	`1` = fraudulent, `0` = legitimate

```
# Import required libraries
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Seaborn style for all plots
sns.set_theme(style='whitegrid')

# Avoid truncation of DataFrame output in Jupyter
pd.set_option('display.expand_frame_repr', False)

# Display all columns in the DataFrame output
pd.set_option('display.max_columns', None)

# Define a separator for better readability
sep = "=" * 50
```

The next two cells load the raw export and clean it as in Notebook 05.

```
# Load the dataset
df = pd.read_csv('vehicle_fraud.csv')
```

```
# Clean as in Notebook 05: remove the duplicates, the placeholder age 0 becomes NaN
df = df.drop_duplicates()
df.loc[df['policyholder_age'] < 18, 'policyholder_age'] = np.nan
```

Phase 1: The Portfolio

Get to know the insured customers and vehicles: ages, customer durations, occupations, regions, vehicle types and coverage types.

Task 1: Plot histograms of `policyholder_age` and `customer_since_years`. Which ages and customer durations are typical?

```
# Task 1: Histograms of age and customer duration
# Hint: .plot(kind='hist', bins=...), labels with plt.title(), plt.xlabel(), plt.ylabel()
#       Whole years: choose bins that each cover the same number of years
```

Task 2: What share of the claims comes from each `occupation` and each `region`?

```
# Task 2: Shares of occupations and regions
# Hint: value_counts(normalize=True)
```

Task 3: Count the claims per `vehicle_type` and `coverage_type` in one table. Which vehicle types are mostly fully insured?

```
# Task 3: Claims per vehicle type and coverage type
# Hint: groupby() with two columns, count(), unstack()
```

Phase 2: Claim Amounts

Compare the claim amounts between groups and find the features with the largest differences.

Claim amounts are right-skewed (Notebook 04), so the median describes a typical claim better than the mean.

Task 1: Compute the median claim amount per `vehicle_type`, `coverage_type` and `region` and plot each as a horizontal bar chart. Which feature shows the largest differences?

```
# Task 1: Median claim amount per vehicle type, coverage type and region
# Hint: groupby()['claim_amount'].median(), sort_values(), plot(kind='barh')
```

Task 2: Create the column `age_group` as in Notebook 03 (`<30`, `30-39`, `40-49`, `50+`). Plot the median claim amount per age group. How does it change with age?

```
# Task 2: Median claim amount per age group
# Hint: pd.cut(..., bins=[0, 30, 40, 50, 100], right=False, labels=[...]),
#       groupby(..., observed=True), plot(kind='line')
```

Task 3: Draw a box plot of `claim_amount` per `coverage_type`. What do the points above the whiskers stand for?

```
# Task 3: Box plot of the claim amounts per coverage type
# Hint: sns.boxplot(data=..., x=..., y=...)
```

Phase 3: Fraud

Find out in which groups fraud is frequent and in which it is not.

`fraud` is `1` for a fraudulent claim, so its mean per group is the fraud rate of the group (Notebook 03).

Task 1: Compute the fraud rate per `vehicle_type`, `coverage_type`, `region` and `occupation`. Does any of these features separate fraudulent from legitimate claims?

```
# Task 1: Fraud rate per vehicle type, coverage type, region and occupation
# Hint: groupby()['fraud'].mean(), a for loop over the four column names
```

Task 2: Plot the fraud rate per age group and per customer duration band: under 5, 5–9, 10–19 and 20 or more years as a customer.

```
# Task 2: Fraud rate per age group and per customer duration band
# Hint: groupby(..., observed=True)['fraud'].mean(), plot(kind='line')
#       pd.cut(df['customer_since_years'], bins=[0, 5, 10, 20, 31], right=False, labels=
    [...])
```

Task 3: Compute the fraud rate per vehicle value band: under 10,000 €, 10,000–20,000 €, 20,000–40,000 € and 40,000 € or more. For which vehicles is fraud most frequent?

```
# Task 3: Fraud rate per vehicle value band
# Hint: pd.cut(df['vehicle_value'], bins=[0, 10000, 20000, 40000, 80000], right=False,
    labels=[...]),
#       groupby(..., observed=True)['fraud'].mean()
```

Conclusion

Task: Summarize your findings in a few sentences: Which features drive the claim amount? Which features point to fraud, and which do not?

Summary

- **Portfolio:** histograms, shares with `value_counts(normalize=True)`, counts per combination with `groupby()` and `unstack()`
- **Claim amounts:** median per group, bar charts, line plots and box plots
- **Fraud:** fraud rate per group, groups from `pd.cut()`

Next: training machine learning models in Notebook 07.